

Amit Paul

SOFTWARE ENGINEER · BACKEND & PLATFORM

Django / DRF · LLM Platform Engineering · Secure Multi-Tenant Systems

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★ CAREER HIGHLIGHTS

- **1,000+ commits, 400+ merged pull requests and 25+ peer reviews** across 17 repositories in three years, spanning Django REST APIs, two forked LLM services, and deployment infrastructure.
- **Enterprise and institutional clients:** UC Berkeley (Kidd Lab), Jazkarta, SmallsLive, and Grameenphone.
- **2,400+ study participants aged 8 to 17** on a COPPA-compliant LLM research platform, where I am top contributor of five engineers (180+ commits).
- **200+ commits on a compliance SaaS:** Stripe billing lifecycle, S3 document management with immutable versioning, and an audit trail on tiered retention.
- **150+ of 266 commits on a learning platform,** built from authentication through the quiz engine to automated certificates.
- **3 NYC jazz venues** on a ticketing platform: fixed 3-D Secure card authentication and per-venue refund and PayPal routing (100+ commits).
- **First author, IEEE ICCIT 2025.** 99.28% accuracy on Vision Transformer bearing-fault diagnosis.

▶ CORE TECHNICAL SKILLS

LANGUAGES	Python (advanced), JavaScript (ES6+), Java, C++, C, SQL, Bash
BACKEND	Django, Django REST Framework, Celery, django-oscar, Node.js / Express, REST API design, drf-yasg / Swagger, WeasyPrint
FRONTEND	Next.js (App Router), React, Vue.js, Nuxt, Tailwind CSS, HTML5, CSS3
DATA & CACHING	PostgreSQL (query planning, N+1 elimination, pg_restore), MongoDB, Redis, Supabase
AUTH & SECURITY	Keycloak (OIDC, PKCE), SimpleJWT, RBAC and permission groups, audit logging, COPPA and UC MSSEI P3
LLM PLATFORM	LiteLLM (fork: guardrails, callbacks, streaming), LibreChat (fork), content moderation guardrails, RAG, OpenAI API
PAYMENTS	Stripe (3-D Secure / SCA, subscriptions, webhooks, refunds), PayPal, SSLCommerz
CLOUD & INFRA	AWS (S3, ECS, CloudWatch), GCP (Cloud Storage, signed URLs), Cloudflare Pages, Docker and Docker Compose, nginx, GitHub Actions
TESTING	pytest, pytest-django, model_bakery, factory fixtures, query-count assertions
CORE	Data Structures & Algorithms, System Design, Performance Optimization, Linux / Unix

📁 PROFESSIONAL EXPERIENCE

Software Engineer — Idlewild Digital, Dhaka, Bangladesh

Mar 2026 – Present

Owns backend and platform delivery, architecture through deployment, across LLM research, compliance, marketplace and ticketing product lines.

- Architected LLM platform integration across three services (Django REST Framework, a forked LiteLLM proxy, a forked LibreChat client), moving chat persistence off middleware hooks into the application request path.
- Engineered age-aware content guardrails (Django rule models, Redis caching, LiteLLM pre- and post-call hooks, OpenAI moderations) including a post-call evaluator that screens generated replies and streams violation events over SSE.
- Built high-volume export pipelines with generator-based streaming CSV, Celery workers, ZIP archives and GCS signed URLs, timezone-correct per user and free of N+1 queries.
- Delivered production payments on Stripe (Payment Intents with 3-D Secure, subscription upgrade and downgrade, twelve webhook handlers, refund-and-rollback reconciliation), plus PayPal and SSLCommerz.

- Designed document management and audit-logging subsystems: S3 behind a provider-agnostic storage layer, immutable versioning enforced by database constraints, checksum deduplication, and CloudWatch logging on tiered retention.
- Enforced multi-tenant authorization with Keycloak SSO (OIDC, PKCE, refresh-and-retry), DRF permission classes and Django group provisioning, to UC Berkeley P3 MSSEI requirements.
- Containerized and deployed Django services on AWS ECS, GCP and VPS (Docker Compose, nginx, GitHub Actions), and traced incidents across layers, including a release script that migrated without granting a new app's permissions.

Backend Developer — Idlewild Digital (Remote)

Jan 2024 – Feb 2026

Django REST Framework delivery for US clients across learning management, freight logistics, community platforms and classifieds.

- Built assessment and certification systems: bulk-write serializer mixins, per-question marking, pass thresholds, progressive content unlocking, and WeasyPrint PDF generation.
- Introduced Celery, Redis, S3 and ffmpeg to an existing codebase, and built media pipelines that extract video duration from uploads and YouTube links and convert images to WebP on save.
- Modelled multi-tenant role systems with rest-framework-roles, custom DRF permission classes, per-verb permission mixins, and database-level uniqueness constraints preceded by data-cleanup migrations.
- Built reporting layers on Django multi-table inheritance with a separate admin site, signal-driven inventory arithmetic that recalculates availability and raises threshold warnings, and eliminated N+1 bottlenecks with select_related, prefetch_related and annotated subqueries.
- Delivered community APIs for forums, articles, webinars, events, connections and reactions behind one reusable owner-or-admin permission base class, with a Next.js frontend integration.

Python Intern — Idlewild Digital

Aug 2023 – Dec 2023

First engineering role. Promoted to Backend Developer in 5 months.

- Owned a Django MVT application end to end: custom user model, authorization mixins, template layer and nginx deployment.
- Extended it with a REST API layer of DRF serializers, generic views, five permission classes composed per action, and SimpleJWT token endpoints.
- Built PDF generation with WeasyPrint including digital signing and QR codes, then automated deployment with a GitHub Actions self-hosted runner.

KEY PROJECTS

- **AYO** COPPA-compliant LLM research platform · UC Berkeley Kidd Lab — Chat client and researcher dashboard for a two-year study of how children aged 8 to 17 talk to LLMs. Four services behind Keycloak SSO: a forked LibreChat client, a forked LiteLLM gateway carrying the child-safety guardrails, a Django/DRF research API, and a Next.js dashboard. *Stack: Django, DRF, PostgreSQL, MongoDB, Redis, Celery, Keycloak, LiteLLM, LibreChat, Next.js*
- **Govara** compliance SaaS — Self-assessment against security and regulatory frameworks. Built the Stripe billing lifecycle, per-tier feature gating, cross-framework evidence propagation, an S3 document library with immutable versioning and rollback, and an audit trail with sensitivity-tiered retention. *Stack: Django, DRF, PostgreSQL, AWS S3, CloudWatch, Celery, Stripe, pytest*
- **ScholarStone** learning platform — Course and assessment platform. Built the quiz engine (submission APIs, per-question marking, a 70% pass mark, unlock gating), automated certificate issuance, the YouTube-backed content pipeline with automatic duration extraction, and dockerized the backend. *Stack: Django REST Framework, PostgreSQL, Celery, Redis, WeasyPrint, ffmpeg, Docker*
- **SmallLive** ticketing for three NYC jazz venues — Stripe Elements checkout with 3-D Secure, per-venue refund and PayPal routing, automatic cancellation on ticket stock-out, protection for reserved tickets when event sets change, and venue-filtered sales reporting. *Stack: Django, django-oscar, PostgreSQL, Stripe, PayPal, Celery*

EDUCATION & PUBLICATIONS

B.Sc. in Electronics & Communication Eng. — Khulna University

Graduated Jan 2026

- **Publication:** Vision Transformer Classification of Bearing Faults, IEEE ICCIT 2025. DOI: 10.1109/ICCIT68739.2025.11490241.
- **Industrial attachment, Grameenphone Ltd. (Jan–Feb 2025):** built a RAG chatbot API for support automation.